Video Transcript

Video 7: Airflow: A Workflow Management Platform (06:57)

When it comes to data platforms, there are many sophisticated arguments that are made by vendors of their platforms. However, when it comes to airflow, this is a pretty accessible pitch. What they're looking at, what they're focusing on is a workflow management platform. And what is a workflow in this case? It's pretty simple, pretty much what it sounds like. A sequence of tasks, something that might launch on a schedule or perhaps on an event, something that is used to build data pipelines. But as you can see, this is really a pretty general problem that applies to many things. So, let's consider what one of these might look like and I'll look here at pulling data, say from transit systems in a city, something like buses, trains, or perhaps ferries. And as you can see here, you start off with raw data and ultimately you have some report or intelligence. You might want to pull that data, do a transformation perhaps to a DOM, extract some tags, from those tags, mine that data, pull out the factors you're looking for and can do a visualization or a report.

Now, there's many ways you could do this. And this is what many of us have done at one point of another over the years, written what's often called glue code, to be able to bring together the different parts that might do that pipeline. You might start off with a bash script that runs on Python to initially pull that data. Then, you might want to do some cleaning, you could do it with Python or some other language. You'd put it into a DOM, if you had some HTML type data, you'd run some JavaScript over that. You'd then pull some tags. You'd do some analytics, perhaps go to a database, use some of the support that is given there, and then ultimately create your visualization or your report. Now to be able to run this periodically, you'd put it into a Chrome job, and over time you'd hit all of those edge cases, all of those problems, and you'd mature and grow it, but it wouldn't come without pain.

There are a lot of common problems that you might run into. One of them would be failures. What if it fails? Do you immediately try again? Do you wait? How long do you wait? On platforms that are URL endpoints, you might be confused or that you might be mistaken for a denial of service. So, you need to be careful with that. You need to be able to monitor to

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capture the amount of times that you have failed, that you have succeeded. You'd eventually want to be able to run some analytics and see how you're doing. There would be also conditional dependencies. What order do you need these tasks to execute? And what if you had a failure? What if you're considering cases where you might get different results by different tasks and you want to do some branching based on that? It also might want to exceed one machine. You might want to have a number of these performing these steps. How do you coordinate between them? You might want to have some deployment considerations, some versioning of your code, and ultimately you might want to rerun some of these historical runs through some of these pipelines.

So, as you can see there, that's a lot of things that are eventually captured by a lot of the homegrown type scripts, but that are challenging to do. And so when it comes to Airflow, this is very much the situation the original author was considering. And you can see there, that a lot of what I've just pointed out was captured on this platform. This is a framework built for that from the ground up. It defines tasks in Python, something that is frequently used in the data community. Something that has the capability to spread throughout different nodes. Something that can capture the history, that has a graphical user interface that allows you to immediately see how this has run, how mature, how successful it uses colors as we will see, that it has an extensible plugin mechanism to a lot of the infrastructure that you'd want to connect.

This is something we didn't even mention, the cycle of trying to find all of the different drivers that you need, the adapters that you need for the different types of databases, and something that has struck an important need when it comes to the community and that has gained wide adoption. Now, the model that it uses to define these tabs or these pipelines are directed acyclic graphs, often called DAGs. And as you can see here, that is a pretty good fit for tasks and the type of sequences that you might want to run at certain points. You might want to have a multitude of these. You might want to have some branching depending on the results of certain intermediate steps. So, why Airflow? For many of the reasons we have mentioned, but I think, ultimately, it comes, to you need it anyway. You need something like it anyway. And so, why not right on top of something that has been built to capture many of the use cases that you're going to run into. And so, this is saving you a great deal of war.

Now, the common use cases are here, I'm listing a few, we've already mentioned some, but ultimately warehouse is something that uses this type of patterns quite a bit, the ETL, extract,

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transform, and load. And, when it comes to machine learning and data science, there's even more of those you can see there and a few others that I'm listing. And, you can see there that I'm also pointing to a URL when you can learn more about this. But if we talk about data science and we talk about the data hierarchy of needs, you can see there that the bottom of the pyramid, the collection, the movement, the transformations are great fits for Airflow.

Now, the rest of the pyramid could make use of it as well. If we were to look at the data science cycle of going from raw data to deployment with data preparation and training in between, you can see there that you're using a multitude of packages. Now, when it comes to the glue that goes between these, you can start to, you could try to do it by hand, as we mentioned before. You could try to use some package that automates a lot of this, but ultimately they're not going to have support for everything that you have. Or you could start to use Airflow. You can see there the different packages and being able to provide that glue might be a pretty useful thing. So, in general, this is something that we need to build anyway and so having a workflow management platform is something that is pretty useful.

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